

Table 1. Spanish-input, English-output TruthfulQA transfer results using English-calibrated controllers. Values are mean  $\pm$  standard error across five repetitions; TBD marks unrun model-method pairs.

|              | Method                      | True (%) $\uparrow$                | Informative (%) $\uparrow$         | Instruction relevance (0–2) $\uparrow$ | Fluency (0–2) $\uparrow$          |
|--------------|-----------------------------|------------------------------------|------------------------------------|----------------------------------------|-----------------------------------|
| Gemma-2-2B   | Original                    | 51.41 $\pm$ 0.90                   | 86.44 $\pm$ 0.58                   | 1.32 $\pm$ 0.02                        | 1.17 $\pm$ 0.00                   |
|              | ITI                         | 53.56 $\pm$ 0.47                   | 88.59 $\pm$ 0.45                   | 1.42 $\pm$ 0.01                        | 1.21 $\pm$ 0.01                   |
|              | ActAdd                      | 63.57 $\pm$ 0.39                   | 67.15 $\pm$ 0.34                   | 0.99 $\pm$ 0.01                        | 0.88 $\pm$ 0.01                   |
|              | Mean-AcT                    | 48.13 $\pm$ 0.52                   | 90.40 $\pm$ 0.46                   | 1.42 $\pm$ 0.01                        | <b>1.23 <math>\pm</math> 0.01</b> |
|              | Linear-AcT                  | 47.83 $\pm$ 0.78                   | 90.60 $\pm$ 0.78                   | <b>1.44 <math>\pm</math> 0.01</b>      | 1.23 $\pm$ 0.01                   |
|              | PID-AcT                     | 48.79 $\pm$ 0.61                   | <b>90.89 <math>\pm</math> 0.49</b> | 1.43 $\pm$ 0.01                        | 1.22 $\pm$ 0.01                   |
|              | ODESteer                    | 57.01 $\pm$ 0.20                   | 86.27 $\pm$ 0.28                   | 1.32 $\pm$ 0.01                        | 1.14 $\pm$ 0.01                   |
|              | S-PID                       | 55.15 $\pm$ 0.37                   | 86.41 $\pm$ 0.34                   | 1.29 $\pm$ 0.01                        | 1.12 $\pm$ 0.00                   |
|              | A-LQR                       | 73.81 $\pm$ 0.71                   | 76.55 $\pm$ 0.65                   | 1.12 $\pm$ 0.01                        | 1.06 $\pm$ 0.00                   |
|              | <b>H<sub>∞</sub> (ours)</b> | <b>77.45 <math>\pm</math> 0.52</b> | 64.16 $\pm$ 0.85                   | 1.04 $\pm$ 0.01                        | 1.02 $\pm$ 0.01                   |
| Llama-3-8B   | Original                    | 42.86 $\pm$ 0.49                   | 94.52 $\pm$ 0.24                   | 1.59 $\pm$ 0.01                        | 1.39 $\pm$ 0.01                   |
|              | ITI                         | 55.47 $\pm$ 0.88                   | 94.37 $\pm$ 0.20                   | 1.61 $\pm$ 0.02                        | 1.39 $\pm$ 0.01                   |
|              | ActAdd                      | 43.94 $\pm$ 0.53                   | 93.78 $\pm$ 0.29                   | 1.57 $\pm$ 0.01                        | 1.35 $\pm$ 0.01                   |
|              | Mean-AcT                    | 46.61 $\pm$ 0.69                   | 94.83 $\pm$ 0.49                   | 1.61 $\pm$ 0.01                        | 1.41 $\pm$ 0.01                   |
|              | Linear-AcT                  | 47.15 $\pm$ 0.41                   | 94.96 $\pm$ 0.39                   | 1.61 $\pm$ 0.01                        | <b>1.42 <math>\pm</math> 0.02</b> |
|              | PID-AcT                     | 47.12 $\pm$ 0.63                   | <b>95.08 <math>\pm</math> 0.51</b> | <b>1.62 <math>\pm</math> 0.01</b>      | 1.41 $\pm$ 0.01                   |
|              | ODESteer                    | <b>95.67 <math>\pm</math> 0.15</b> | 11.21 $\pm$ 0.45                   | 0.00 $\pm$ 0.00                        | 0.02 $\pm$ 0.00                   |
|              | S-PID                       | 52.09 $\pm$ 0.56                   | 94.37 $\pm$ 0.22                   | 1.59 $\pm$ 0.01                        | 1.33 $\pm$ 0.01                   |
|              | A-LQR                       | 55.57 $\pm$ 0.71                   | 95.03 $\pm$ 0.25                   | 1.56 $\pm$ 0.01                        | 1.27 $\pm$ 0.01                   |
|              | <b>H<sub>∞</sub> (ours)</b> | 85.21 $\pm$ 0.40                   | 64.70 $\pm$ 0.48                   | 0.97 $\pm$ 0.03                        | 0.97 $\pm$ 0.02                   |
| Qwen-2.5-14B | Original                    | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | ITI                         | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | ActAdd                      | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | Mean-AcT                    | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | Linear-AcT                  | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | PID-AcT                     | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | ODESteer                    | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | S-PID                       | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | A-LQR                       | TBD                                | TBD                                | TBD                                    | TBD                               |
|              | <b>H<sub>∞</sub> (ours)</b> | TBD                                | TBD                                | TBD                                    | TBD                               |